

JITHIN JOSEPH

AI ENGINEER

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PROFESSIONAL SUMMARY

AI Engineer who independently designed, built, and deployed a multimodal Retrieval-Augmented Generation (RAG) pipeline — from cross-modal embedding architecture through to a live, publicly accessible application on Streamlit Community Cloud. Hands-on with vector search architecture (Qdrant, FAISS, ChromaDB), cross-modal retrieval combining vision and speech models, and full-stack delivery with Django and FastAPI. Comfortable owning a project end-to-end — architecture, model integration, evaluation, and deployment — without relying on a team to carry any single piece. Brings a research-driven, detail-critical mindset from two years in a compliance-heavy healthcare lab, now applied to building, evaluating, and shipping AI systems end-to-end.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML5, CSS3

RAG & GenAI: Retrieval-Augmented Generation (naive, modular, agentic), Multimodal RAG, Prompt Engineering, Context Grounding, LLM Pipelines, Vector Search

Vision & Speech Models: CLIP (image embeddings), Sentence-Transformers/BGE (text embeddings), YOLOv8 / YOLOv8n (object detection), faster-whisper (speech-to-text), Tesseract OCR

Generative & ML Models: SDXL-Turbo, Stable Diffusion, Groq Llama-3.3-70B, Google Gemini, RandomForest, XGBoost, IsolationForest, LangChain, Hugging Face Transformers, OpenAI API

Vector Databases: Qdrant (multi-collection retrieval), ChromaDB, FAISS

Backend & Deployment: Django, FastAPI, RESTful API Design, Streamlit Community Cloud, Git/GitHub, Docker

Data Processing & Tools: PyPDF, python-docx, python-pptx, pandas, Pillow, ffmpeg, MySQL, SQLite, Postman, VS Code

PROJECTS

Infinity — Multi modal RAG System — Personal project — deployed, local-first RAG pipeline — [GitHub](#)

- Architecture: a modular pipeline (Ingestion → Embeddings → Retrieval → Generation, each an isolated module) accepts 27+ file types across documents, images, audio, and video, extracts and chunks content, indexes it into two dedicated Qdrant collections, and answers questions grounded strictly in retrieved context.
- Embeddings: Sentence-Transformer/BGE text embeddings and CLIP image embeddings, indexed into two separate Qdrant collections (rag_text, rag_image) so text and visual search can run independently or be combined at query time.
- Vision, OCR & speech: YOLOv8n for video-frame object detection, Tesseract OCR and vision-model description for images/frames, and faster-whisper for audio/video transcription — all feeding a unified text index.
- Generation: answer synthesis via Groq and Google Gemini, with responses grounded strictly in retrieved context to reduce hallucination — deployed end-to-end as a live Streamlit application on Streamlit Community Cloud.

Atelier AI — Architect Assistant — Personal project — generative design studio — [GitHub](#)

- Architecture: a Streamlit front end drives a three-stage workflow — Brief, Plan Studio, Visualization — with session-state management carrying the user's brief and requirements forward into each generation stage.
- Model layer: SDXL-Turbo via Hugging Face Diffusers, with automatic device selection (CUDA → Apple MPS → CPU fallback) and dynamic precision switching (fp16 on GPU, fp32 on CPU) to keep generation fast across hardware.
- Differentiated prompt engineering per stage: blueprint-style prompts (top-down, black-and-white, technical) for floor plans versus photorealistic prompts (24mm architectural photography, natural materials, lighting) for final renders, generating 3 parallel options at each stage.
- Tuned inference steps and guidance scale to trade off generation speed against output quality, and built a custom dark-themed UI (blueprint-grid background, drafting-sheet panels) rather than using Streamlit's default theme.

RoadAssist — Roadside Assistance & Emergency Fuel Service — Personal project — Django full-stack web app — [GitHub](#)

- Architecture: a Django application with a custom user model (AbstractUser subclass) supporting two roles — user and worker — with role-based redirect on login routing each to a dedicated dashboard.
- Modeled two independent request workflows — fuel delivery (type, quantity, unit, location) and roadside service (vehicle type, issue category, optional image upload via Pillow) — each with its own status lifecycle (pending → in progress → completed) and worker assignment.
- Designed ModelForms that deliberately exclude status and assigned-worker fields from user input, keeping request-state transitions controlled by application logic rather than the client — and registered both request types in Django admin with custom admin classes for staff-side management.

PROFESSIONAL EXPERIENCE

Lab Coordinator

Oct 2023 – Oct 2025

MVR Cancer Center and Research Institute, Calicut

- Coordinated day-to-day laboratory operations and cross-departmental communication in a compliance-driven healthcare research environment, standardizing documentation practices that reduced handoff delays.
- Developed a structured, detail-critical approach to process documentation and quality control — habits now applied to debugging and validating AI system outputs.

EDUCATION

BSc Honours – Psychology

Jun 2020 – Jun 2023

St. Alphonsa College, Calicut University

CERTIFICATIONS

Python Full Stack Development with Gen AI

Jan 2026 –

Present

Haris and Co. Tech School, Calicut

ADDITIONAL INFORMATION

Languages Spoken: English, Malayalam, Tamil, Hindi **Volunteering:** Active Member, Junior Red Cross